

DSA Sheet 1 - 20 Problems

--- Problem 1: Sum of Array Elements ---

Problem Statement

You are given an array of integers. Your task is to calculate the sum of all elements present in the array.

This is a basic operation frequently used in programming and helps in building a strong foundation for solving more complex problems.

Input: An array of integers.

Output: A single integer representing the sum of all elements in the array.

Input Format

The first line contains an integer n - the number of elements in the array.

The second line contains n space-separated integers representing the array elements.

Output Format

Print a single integer - the sum of all elements in the array.

Example 1

Input:

6

1 2 3 4 5 6

Output: 21

Explanation: $1 + 2 + 3 + 4 + 5 + 6 = 21$

Example 2

Input:

3

5 10 15

Output: 30

Explanation: $5 + 10 + 15 = 30$

--- Problem 2: Linear Search ---

Problem Statement

You are given an array of integers `arr[]` of size n and a target element x . Your task is to search for the element x in the array.

If the element is present, return its index (0-based indexing). If the element is not found, return -1.

This problem is based on the Linear Search technique, where each element is checked one by one.

Input: An array of integers; A target integer x

Output: An integer representing the index of element x if found; Otherwise, -1

Input Format

The first line contains an integer n - number of elements in the array

The second line contains n space-separated integers

The third line contains an integer x - the element to search

Output Format

Print a single integer: Index of x (0-based), if present; -1 if not present

Example 1

Input:

10

10 20 30 40 50 60 70 80 90 100

70

Output: 6

Explanation: Element 70 is found at index 6 (0-based indexing).

Example 2

Input:

9

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1 2 3 4 5 6 7 8 9

6

Output: 5

Explanation: Element 6 is found at index 5.

Example 3 (Not Found Case)

Input:

5

2 4 6 8 10

7

Output: -1

Explanation: Element 7 is not present in the array.

--- Problem 3: Contains Duplicate ---

Problem Statement

You are given an integer array `nums`. Your task is to determine whether any value appears at least twice in the array.

If any element is repeated, return `true`. If all elements are distinct (unique), return `false`.

This problem helps in understanding basic array traversal and duplicate detection.

Input: An array of integers.

Output: Return `true` if any value appears at least twice; Return `false` if all elements are distinct

Input Format

The first line contains an integer `n` - number of elements in the array

The second line contains `n` space-separated integers representing the array

Output Format

Print `true` or `false`

Example 1

Input:

6

1 2 1 3 4 5

Output: `true`

Explanation: The element 1 appears more than once (at index 0 and 2), so the output is `true`.

Example 2

Input:

7

1 2 3 4 5 6 7

Output: `false`

Explanation: All elements are unique, so the output is `false`.

--- Problem 4: Two Sum ---

Problem Statement

You are given an array of integers `nums` and an integer `target`. Your task is to find two distinct indices such that the numbers at those indices add up to the target.

Return the indices (0-based) of the two numbers. If no such pair exists, return -1 (or an empty result, depending on requirement).

This is a classic Two Sum problem.

Input: An array of integers `nums`; An integer `target`

Output: A pair of indices `[i, j]` such that `nums[i] + nums[j] = target`. If no such pair exists, return -1

Input Format

The first line contains an integer `n` - number of elements in the array

The second line contains `n` space-separated integers

The third line contains an integer `target`

Output Format

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Print two space-separated integers representing the indices. If no solution exists, print -1

Example 1

Input:

4

2 7 11 15

9

Output: 0 1

Explanation: $\text{nums}[0] = 2$ and $\text{nums}[1] = 7$; $2 + 7 = 9$ matches target

Example 2

Input:

4

2 7 11 15

18

Output: 1 2

Explanation: $\text{nums}[1] = 7$ and $\text{nums}[2] = 11$; $7 + 11 = 18$

Example 3 (No Solution)

Input:

4

1 2 3 4

10

Output: -1

Explanation: No two numbers sum to 10.

--- Problem 5: Intersection of Two Arrays ---

Problem Statement

You are given two integer arrays nums1 and nums2 . Your task is to return an array representing their intersection. Each element in the result must be unique. The result can be returned in any order.

Input: Two arrays of integers: nums1 and nums2

Output: An array containing unique common elements present in both arrays

Input Format

The first line contains an integer n - number of elements in nums1

The second line contains n space-separated integers

The third line contains an integer m - number of elements in nums2

The fourth line contains m space-separated integers

Output Format

Print the array containing unique intersection elements (space-separated or in list format)

Example 1

Input:

3

4 9 5

5

9 4 9 8 4

Output: 4 9

Explanation: Common elements are 4 and 9. Even though they appear multiple times, we include them only once.

Example 2

Input:

6

1 2 3 4 5 6

5

2 4 5 2 9

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Output: 2 4 5

Explanation: Common unique elements are 2, 4, and 5.

--- Problem 6: Maximum and Minimum in Array ---

Problem Statement

You are given an array of integers `arr[]` of size `N`. Your task is to find the maximum and minimum elements in the array. The maximum element is the largest value in the array. The minimum element is the smallest value in the array.

Input: An array of integers.

Output: Print the maximum element; Print the minimum element

Input Format

The first line contains an integer `N` - number of elements in the array

The second line contains `N` space-separated integers

Output Format

Print two lines:

Maximum Number is: <value>

Minimum Number is: <value>

Example 1

Input:

9

1 2 3 4 5 6 7 8 9

Output:

Maximum Number is: 9

Minimum Number is: 1

Explanation: The largest element is 9 and the smallest element is 1.

Example 2

Input:

5

6 7 18 3 9

Output:

Maximum Number is: 18

Minimum Number is: 3

Explanation: The largest element is 18 and the smallest is 3.

--- Problem 7: Perfect Square ---

Problem Statement

You are given a positive integer `num`. Your task is to determine whether the number is a perfect square.

A number is called a perfect square if it can be expressed as the product of an integer multiplied by itself (i.e., $x * x = num$).

Return true if `num` is a perfect square; Otherwise, return false

Input: A single positive integer `num`

Output: Print true if the number is a perfect square; Print false otherwise

Input Format: A single integer `num`

Output Format: Print true or false

Example 1

Input: 81

Output: true

Explanation: $81 = 9 * 9$ - perfect square

Example 2

Input: 64

Output: true

Explanation: $64 = 8 * 8$ - perfect square

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Example 3

Input: 10

Output: false

Explanation: 10 cannot be expressed as $x * x$ for any integer x

--- Problem 8: Factorial ---

Problem Statement

You are given a positive integer n . Your task is to calculate the factorial of the number.

The factorial of a number n is defined as: $n! = n * (n-1) * (n-2) * \dots * 1$

Input: A single positive integer n

Output: A single integer representing the factorial of n

Input Format: A single integer n

Output Format: Print the value of $n!$

Example 1

Input: 5

Output: 120

Explanation: $5! = 5 * 4 * 3 * 2 * 1 = 120$

Example 2

Input: 6

Output: 720

Explanation: $6! = 6 * 5 * 4 * 3 * 2 * 1 = 720$

--- Problem 9: Armstrong Number ---

Problem Statement

You are given a positive integer num . Your task is to determine whether the number is an Armstrong number.

An Armstrong number (also called a Narcissistic number) is a number in which the sum of its digits each raised to the power of the number of digits is equal to the number itself.

Input: A single positive integer num

Output: Print true if the number is an Armstrong number; Print false otherwise

Input Format: A single integer num

Output Format: Print true or false

Example 1

Input: 153

Output: true

Explanation: Number of digits = 3; $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$. So, it is an Armstrong number.

Example 2

Input: 370

Output: true

Explanation: Number of digits = 3; $3^3 + 7^3 + 0^3 = 27 + 343 + 0 = 370$.

Example 3

Input: 123

Output: false

Explanation: $1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36 \neq 123$.

--- Problem 10: Subarrays with Sum Equal to Target ---

Problem Statement

You are given an integer array `arr[]` and an integer target. Your task is to find all continuous subarrays whose sum is equal to the given target.

A subarray is a sequence of elements that are contiguous (next to each other) in the array.

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Input: An array of integers `arr[]`; An integer `target`

Output: Print all subarrays whose sum equals the target. Each subarray should be printed separately.

Input Format

The first line contains an integer `n` - number of elements in the array

The second line contains `n` space-separated integers

The third line contains an integer `target`

Output Format

Print each valid subarray on a new line

Example 1

Input:

7

2 5 -3 1 4 -2 3

6

Output:

2 5 -3 1 4 -2

5 -3 1 4 -2 3

1 4 -2 3

3 1 4 -2

Explanation: The following subarrays have sum = 6:

[2, 5, -3, 1, 4, -2], [5, -3, 1, 4, -2, 3], [1, 4, -2, 3], [3, 1, 4, -2]

All are continuous parts of the array.

--- Problem 11: Missing Number ---

Problem Statement

You are given an array `nums[]` containing `n` distinct numbers taken from the range `[0, n]`. Your task is to find the only number missing from this range.

Constraints

`n == nums.length`

`1 <= n <= 10^4`

`0 <= nums[i] <= n`

All elements in `nums` are unique

Input: An array of integers `nums[]`

Output: A single integer representing the missing number

Input Format

The first line contains an integer `n` - number of elements

The second line contains `n` space-separated integers

Output Format

Print the missing number

Example 1

Input:

3

3 0 1

Output: 2

Explanation: Numbers should be from 0 to 3 -> {0, 1, 2, 3}. But 2 is missing.

Example 2

Input:

4

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0 1 2 3

Output: 4

Explanation: Numbers should be from 0 to 4 -> {0, 1, 2, 3, 4}. But 4 is missing.

--- Problem 12: Sum of Square Numbers ---

Problem Statement

You are given a non-negative integer c . Your task is to determine whether there exist two integers a and b such that:

$$a^2 + b^2 = c$$

Return true if such integers exist; Otherwise, return false

Constraints: $0 \leq c \leq 2^{31} - 1$

Input: A single non-negative integer c

Output: Print true if there exist integers a and b satisfying the condition; Print false otherwise

Input Format: A single integer c

Output Format: Print true or false

Example 1

Input: 5

Output: true

Explanation: We can take $a = 1$, $b = 2$; $1^2 + 2^2 = 1 + 4 = 5$

Example 2

Input: 3

Output: false

Explanation: There are no integers a and b such that $a^2 + b^2 = 3$

--- Problem 13: Valid Anagram ---

Problem Statement

You are given two strings s and t . Your task is to determine whether t is an anagram of s .

An anagram means both strings contain: The same characters; With the same frequency; But possibly in a different order.

Return true if t is an anagram of s ; Otherwise, return false

Constraints: $1 \leq s.length, t.length \leq 5 \cdot 10^4$; s and t consist of lowercase English letters

Input: Two strings s and t

Output: Print true if t is an anagram of s ; Print false otherwise

Input Format: First line contains string s ; Second line contains string t

Output Format: Print true or false

Example 1

Input:

anagram

nagaram

Output: true

Explanation: Both strings contain the same characters with the same frequency.

Example 2

Input:

rat

car

Output: false

Explanation: Characters and frequencies do not match.

--- Problem 14: Isomorphic Strings ---

Problem Statement

You are given two strings s and t . Your task is to determine whether the two strings are isomorphic.

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Two strings are isomorphic if:

Each character in s can be mapped to a character in t

The mapping is consistent

No two characters in s map to the same character in t

A character may map to itself

The order of characters must be preserved

Return true if the strings are isomorphic; Otherwise, return false

Constraints: $1 \leq s.length \leq 5 * 10^4$; $t.length == s.length$; s and t consist of valid ASCII characters

Input: Two strings s and t

Output: Print true if the strings are isomorphic; Print false otherwise

Input Format: First line contains string s; Second line contains string t

Output Format: Print true or false

Example 1

Input:

egg

add

Output: true

Explanation: e -> a, g -> d. Mapping is consistent and valid.

Example 2

Input:

foo

bar

Output: false

Explanation: o maps to two different characters - invalid mapping.

--- Problem 15: String to Integer (atoi) ---

Problem Statement

Implement a function `myAtoi(string s)` that converts a given string into a 32-bit signed integer.

The conversion follows these rules:

Ignore leading whitespaces

Check sign ('+' or '-')

Default is positive if no sign is present

Read digits until a non-digit character is found

If no digits are found, return 0

Clamp the result within the 32-bit signed integer range:

Minimum: -2^{31}

Maximum: $2^{31} - 1$

Constraints: $0 \leq s.length \leq 200$; s consists of English letters, digits, '+', '-', '.', and whitespace

Input: A string s

Output: A 32-bit signed integer after conversion

Input Format: A single string s

Output Format: Print the converted integer

Example 1: Input 42 -> Output 42

Example 2: Input " -42" -> Output -42 (leading spaces ignored, negative sign)

Example 3: Input "4193 with words" -> Output 4193 (stops at non-digit)

Example 4: Input "words and 987" -> Output 0 (no valid number at start)

Example 5 (Overflow): Input 91283472332 -> Output 2147483647 (clamped to $2^{31} - 1$)

--- Problem 16: Second Smallest and Second Largest ---

Problem Statement

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You are given an array of integers `arr[]`. Your task is to find:

The second smallest element; The second largest element

If duplicate values exist, consider distinct elements only while determining the result.

Input: An array of integers `arr[]`

Output: Print the second smallest element; Print the second largest element

Input Format

The first line contains an integer `n` - number of elements in the array

The second line contains `n` space-separated integers

Output Format

Print two lines:

Second Smallest: <value>

Second Largest: <value>

Example 1

Input:

5

3 1 5 2 4

Output:

Second Smallest: 2

Second Largest: 4

Explanation: Sorted distinct array -> {1, 2, 3, 4, 5}; Second smallest = 2; Second largest = 4

Example 2

Input:

5

10 10 20 20 30

Output:

Second Smallest: 20

Second Largest: 20

Explanation: Distinct elements -> {10, 20, 30}; Second smallest = 20; Second largest = 20

--- Problem 17: Fibonacci Number ---

Problem Statement

The Fibonacci sequence is a series of numbers where each number is the sum of the two preceding ones, starting from 0 and 1.

$F(0) = 0$, $F(1) = 1$

$F(n) = F(n-1) + F(n-2)$, for $n > 1$

You are given an integer `n`. Your task is to compute the `n`-th Fibonacci number.

Constraints: $0 \leq n \leq 30$

Input: A single integer `n`

Output: Print the value of the `n`-th Fibonacci number

Input Format: A single integer `n`

Output Format: Print `F(n)`

Example 1

Input: 2

Output: 1

Explanation: $F(2) = F(1) + F(0) = 1 + 0 = 1$

Example 2

Input: 10

Output: 55

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Explanation: Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55. So $F(10) = 55$

--- Problem 18: Pow(x, n) ---

Problem Statement

Implement a function `pow(x, n)` that calculates the value of a number x raised to the power n , i.e., x^n

Constraints:

$-100.0 < x < 100.0$

$-2^{31} \leq n \leq 2^{31} - 1$

n is an integer

Either $x \neq 0$ or $n > 0$

$-10^4 \leq x^n \leq 10^4$

Input: A floating-point number x ; An integer n

Output: A floating-point number representing x^n

Input Format: First line contains a floating-point number x ; Second line contains an integer n

Output Format: Print the result of x^n

Example 1

Input:

2.00000

10

Output: 1024.00000

Explanation: $2^{10} = 1024$

Example 2

Input:

2.10000

3

Output: 9.26100

Explanation: $2.1^3 = 2.1 * 2.1 * 2.1 = 9.261$

Example 3

Input:

2.00000

-2

Output: 0.25000

Explanation: $2^{(-2)} = 1 / (2^2) = 1/4 = 0.25$

--- Problem 19: Search a 2D Matrix ---

Problem Statement

You are given an $m \times n$ integer matrix `matrix` with the following properties:

Each row is sorted in non-decreasing order

The first element of each row is greater than the last element of the previous row

You are also given an integer `target`. Your task is to determine whether the target exists in the matrix.

You must solve the problem with time complexity $O(\log(m * n))$.

Return true if the target exists; Otherwise, return false

Constraints:

$m == \text{matrix.length}$

$n == \text{matrix}[i].length$

$1 \leq m, n \leq 100$

$-10^4 \leq \text{matrix}[i][j], \text{target} \leq 10^4$

Input: A 2D array (`matrix`) of integers; An integer `target`

Output: Print true if target is found; Print false otherwise

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Input Format

First line contains two integers m and n

Next m lines each contain n space-separated integers

Last line contains integer target

Output Format: Print true or false

Example 1

Input:

3 4

1 3 5 7

10 11 16 20

23 30 34 60

3

Output: true

Explanation: The element 3 is present in the matrix.

Example 2

Input:

3 4

1 3 5 7

10 11 16 20

23 30 34 60

13

Output: false

Explanation: The element 13 is not present in the matrix.

--- Problem 20: Power of Three ---

Problem Statement

You are given an integer n. Your task is to determine whether the number is a power of three.

A number is a power of three if there exists an integer x such that: $n = 3^x$

Return true if n is a power of three; Otherwise, return false

Constraints: $-2^{31} \leq n \leq 2^{31} - 1$

Input: A single integer n

Output: Print true if n is a power of three; Print false otherwise

Input Format: A single integer n

Output Format: Print true or false

Example 1

Input: 27

Output: true

Explanation: $27 = 3^3$ - power of three

Example 2

Input: 0

Output: false

Explanation: 0 cannot be expressed as 3 raised to any power.

Example 3

Input: 9

Output: true

Explanation: $9 = 3^2$ - power of three